

New Graph Layouts for GRNsight:

a web application for visualizing models of gene regulatory networks

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LMU|LA
Frank R. Seaver College
of Science and Engineering

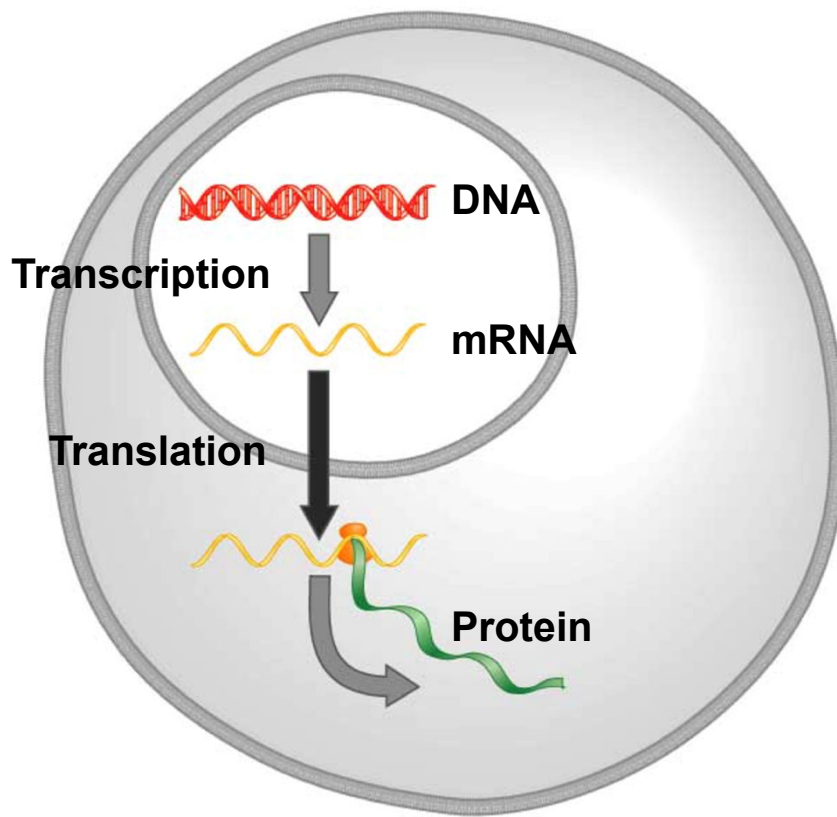
Outline

- **Gene regulatory networks (GRNs) consist of genes, transcription factors, and the regulatory relationships between them.**
- **GRNsight is an open source web application and service for visualizing models of gene regulatory networks.**
- **In GRNsight version 3.0, grid layout function was implemented.**

Outline

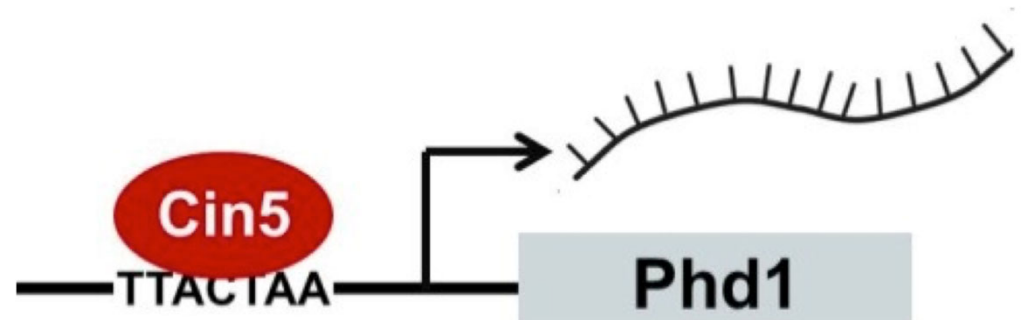
- **Gene regulatory networks (GRNs) consist of genes, transcription factors, and the regulatory relationships between them.**
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A gene regulatory network (GRN) controls the level of expression of genes and proteins



Gene regulatory process
(Freeman, 2002)

- Transcription factors **activate** or **repress** expression.
- To understand GRNs we want to quantify the **weight values** of these regulatory relationships



GRNmap models the dynamics of GRNs to estimate weight parameters (w)



- Magnitude of weights represent the influence of the transcription factor on its target gene.
- **Positive** weights = **activation** of gene expression.
- **Negative** weights = **repression** of gene expression.

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\left(\sum_j (w_{ij}x_j(t)) - b_i\right)\right)} - d_i x_i(t)$$

GRNmap differential equation

GRNmap outputs Excel spreadsheets, with estimated weight parameters

	A
1	cols regulators/rows target
2	ABF1
3	ACE2
4	AFT1
5	CIN5
6	CUP9
7	FHL1
8	GTS1
9	HAL9
10	HSF1
11	MAC1
12	MSN1
13	MSN4
14	NRG1
15	PHD1
16	RAP1
17	REB1
18	ROX1
19	RPH1
20	SKN7
21	YAP1
22	YAP6

.xlsx

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    </node>
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```

.graphml

CIN5	0.9393	MSN1
CIN5	-0.9278	ROX1
CIN5	-0.5312	YAP6
CUP9	-0.1293	YAP6
FHL1		
GTS1		
HSF1	-0.0102	REB1
MSN1		
MSN4	0.6121	FHL1
NRG1	1.2341	NRG1
NRG1	0.6215	YAP6
RAP1	-1.2321	HSF1
RAP1	1.0131	MSN4
RAP1	-0.8890	RAP1
RAP1	-0.4030	AFT1
RAP1	1.4999	RPH1
AFT1	-0.8966	AFT1
REB1	0.0778	GTS1
ROX1	-0.7503	YAP6
RPH1		
YAP1	-0.4315	ROX1
YAP1	0.0146	YAP6
YAP6	-0.0450	CIN5
YAP6	-0.5071	ROX1
YAP6	-0.3027	YAP6
ABF1	0.1562	FHL1
ABF1	-2.9707	MSN1
ACE2	-1.3615	YAP1
HAL9	1.4283	MSN4
MAC1	-0.1882	CUP9
PHD1	-0.6510	CUP9
PHD1	0.5447	MSN4
SKN7	-0.1852	NRG1
SKN7	0.5744	ROX1
SKN7	-0.4082	YAP1

.sif

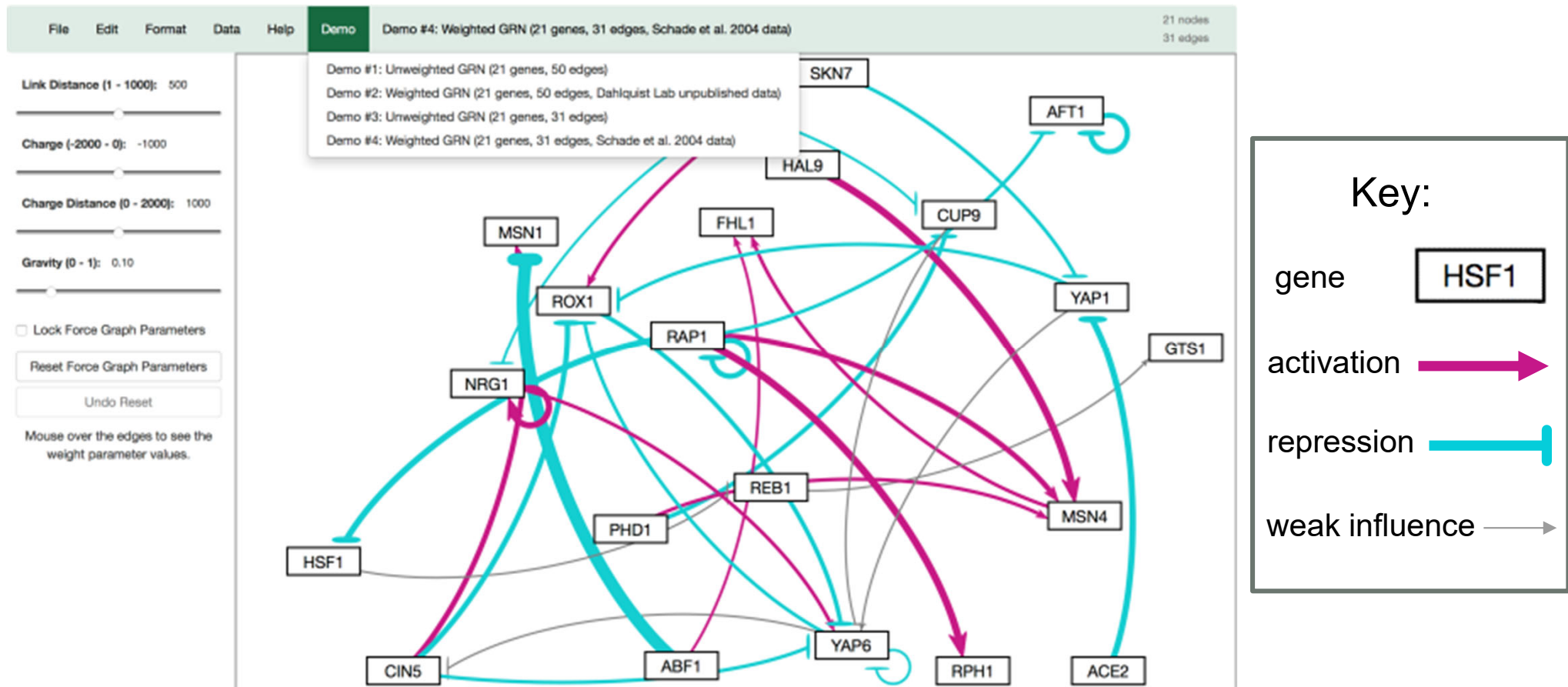
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DX1	RPH1	SKN7	YAP1	YAP6
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	0	0	0	-0.0450
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	-0.1852	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0.5744	-0.4315	-0.5071
	0	0	0	0
	0	0	0	0
	0	-0.4082	0	0
0.7503	0	0	0.0146	-0.3027

21 gene, 31 edge weighted network output from GRNmap

Outline

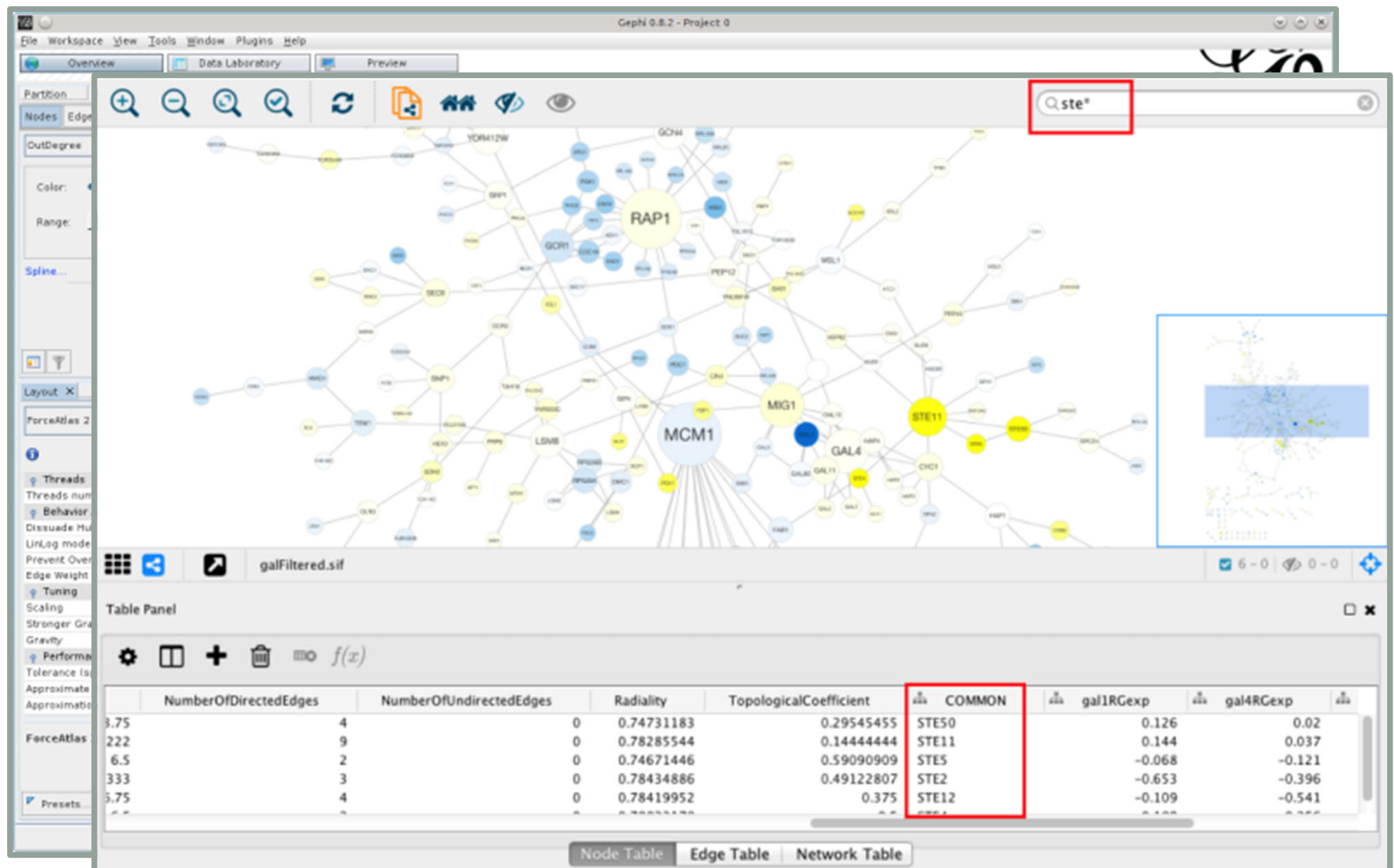
- Gene regulatory networks (GRNs) consist of genes, transcription factors, and the regulatory relationships between them.
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- In GRNsight version 3.0, grid layout function was implemented.

GRNsight automatically produces visualizations of weighted networks from adjacency matrix inputs



Screenshot of GRNsight v1, released Aug 2016

Existing visualization software must be installed and is optimized for large-scale graphs



Cytoscape – cytoscape.org

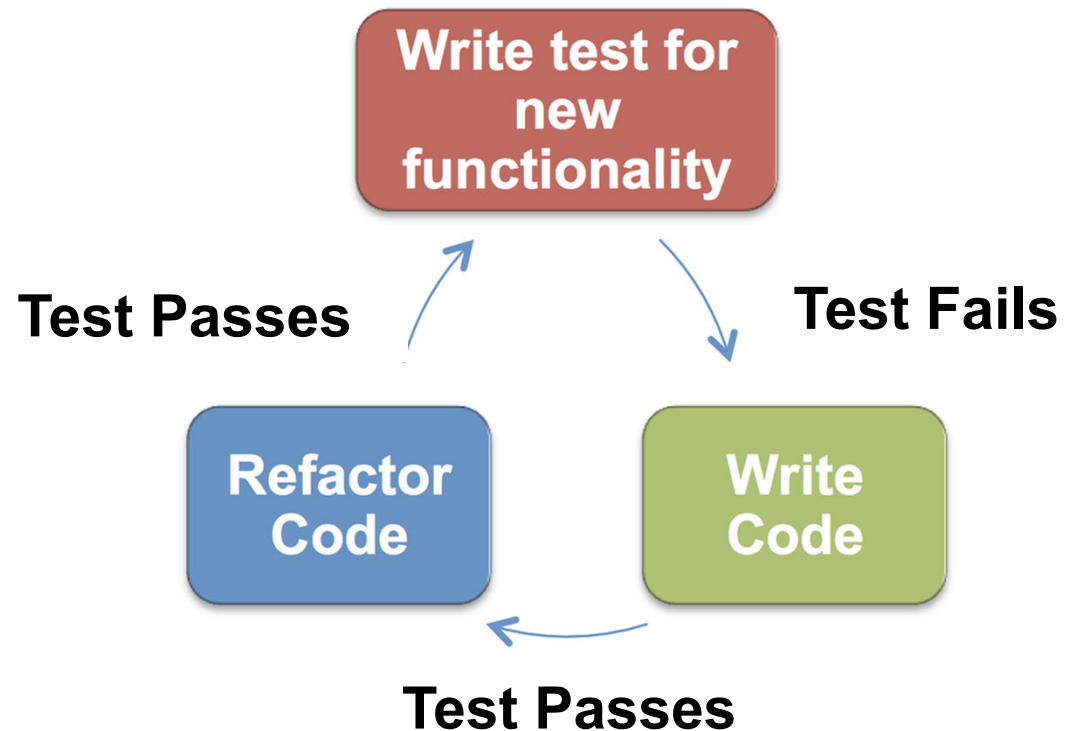
GRNsight fills a niche in bioinformatics visualization software

Cytoscape and Gephi	GRNsight
Optimized for large-scale graphs.	✓ Optimized for small- to medium-scale graphs.
Must be installed.	✓ Exists as a web application.
Several complex features in addition to visualization.	✓ Easy to use, and easily interpretable visualizations.

GRNsight is written in JavaScript and follows software engineering best practices



**GRNsight
technology stack**

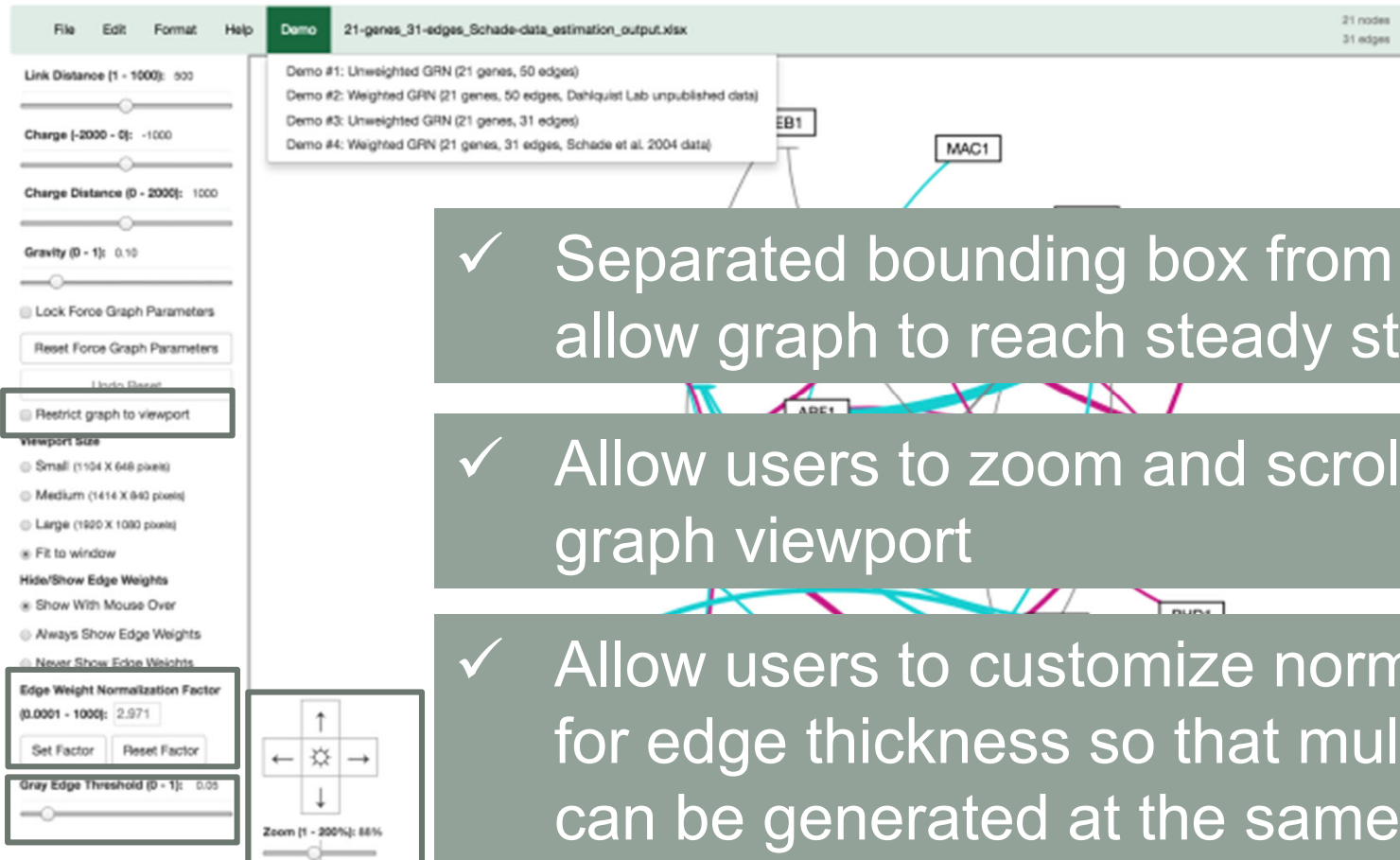


**Test-driven
development cycle**

GRNsight v2 (beta) Screenshot

Beta v2.2.3

Disclaimer: This is the beta version of GRNsight for live testing of new functionality. For the most stable version, please go to the GRNsight home page.



✓ Separated bounding box from the viewport to allow graph to reach steady state

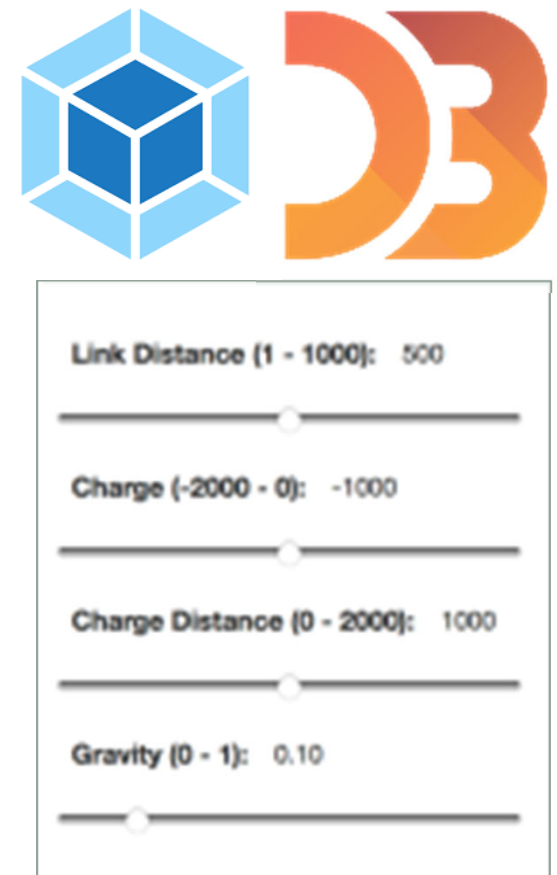
✓ Allow users to zoom and scroll within the graph viewport

✓ Allow users to customize normalization factor for edge thickness so that multiple graphs can be generated at the same scale

✓ Allow users to customize gray edge threshold so that a user can decide the threshold to determining if a relationship is weak

GRNsight changes from v.2 to v.3

- Refactored GRNsight code using D3 library to v.4 from v3.
 - Simple changes with keyword replacement
 - Zooming
 - Force Simulation
- Two force sliders removed
 - Charge Distance
 - Gravity
- **Node coloring** by Eileen Choe 2018
- **Grid Layout** by Jen Shin 2018
 - Included Webpack in the package to import and export methods and objects within GRNsight files



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Grid layout feature developed for GRNsight v3 is highlighted in this presentation

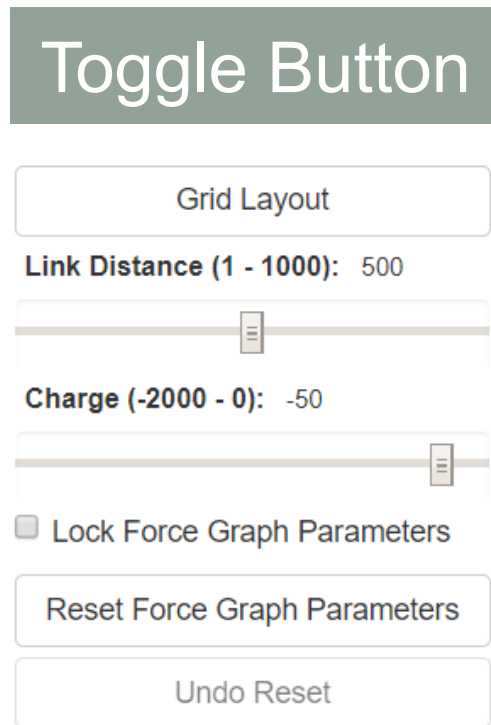
Corresponding Objectives:

1. Allow the user to toggle between force graph and grid layout
2. Organize the nodes in a grid pattern
3. Sort the nodes in alphabetical order

Full list of v3 features documented in the release notes:

<http://dondi.github.io/GRNsight/about.html#releaseContent>

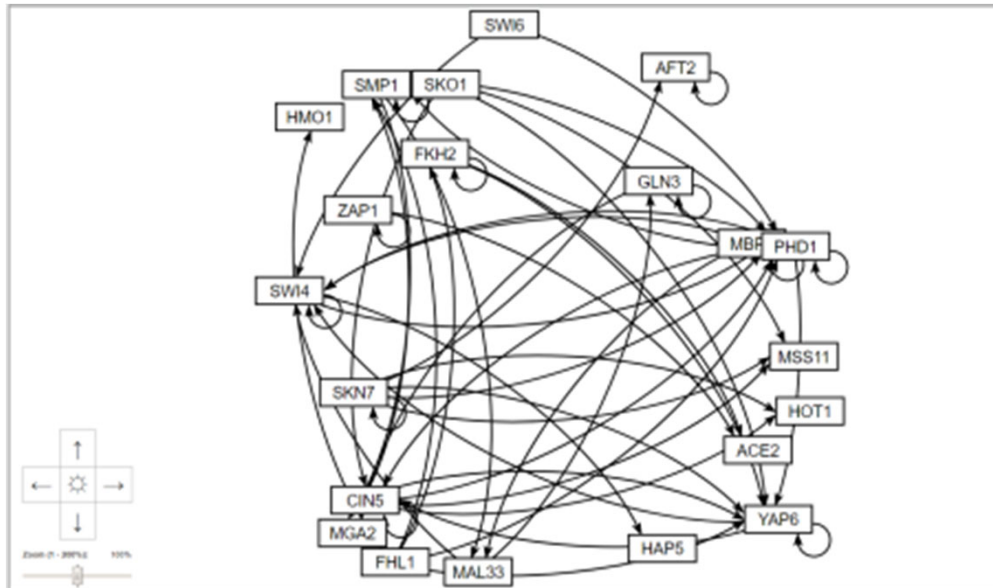
Allow the user to toggle between force graph and grid layout



- Users can easily apply grid layout by clicking the button on the left menu bar.
- Users can switch back to the force graph layout by clicking the grid layout button again.
- Toggling back and forth triggers a graph reload.
- Future plans to reorganize the menu in order to display the grid layout and force graph layout options more distinctively.

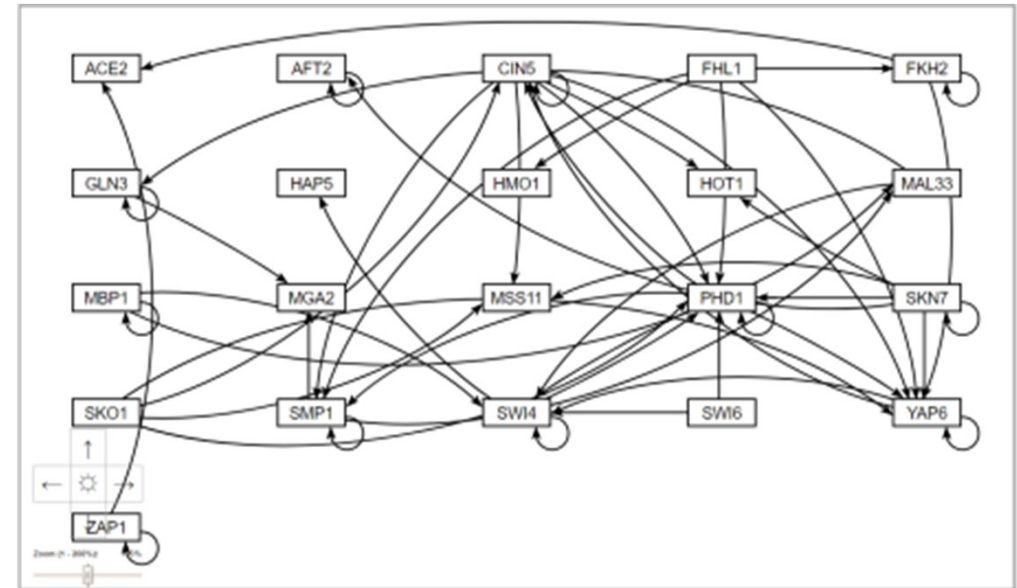
Allow the user to toggle between force graph and grid layout

Force Graph



- Initial state when graph is first loaded.
- Nodes float and move with applied force.

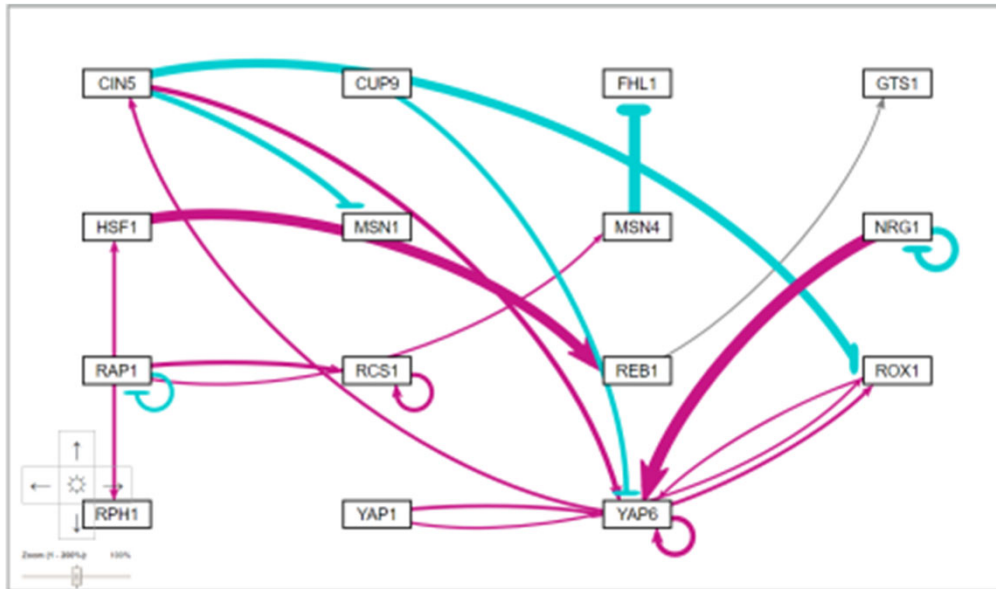
Grid Layout



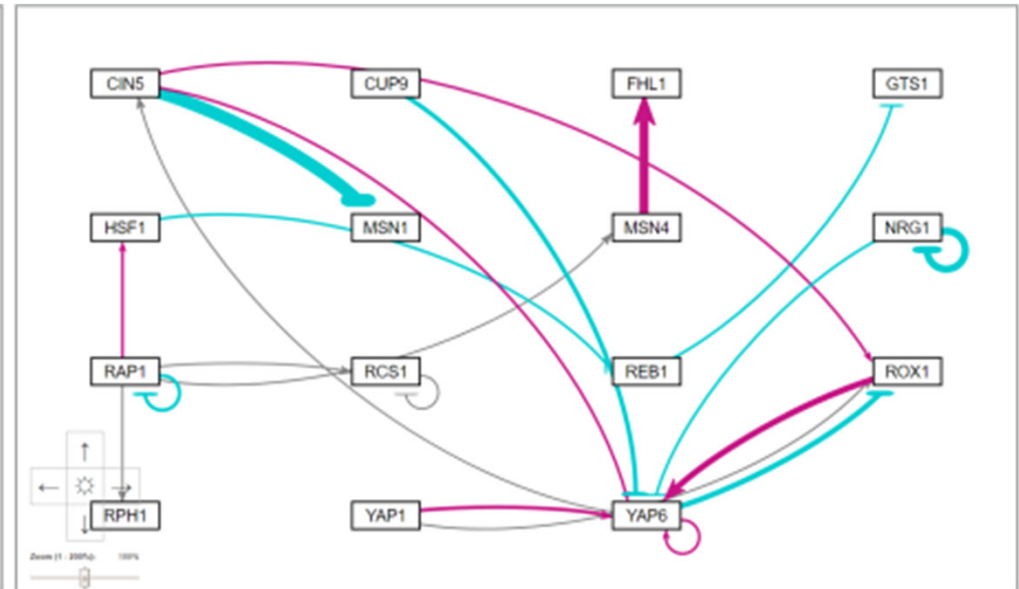
- Nodes do not float like in force graph.
- Users can move the nodes around like in force graph.

Organize the nodes in a grid pattern

test-data1



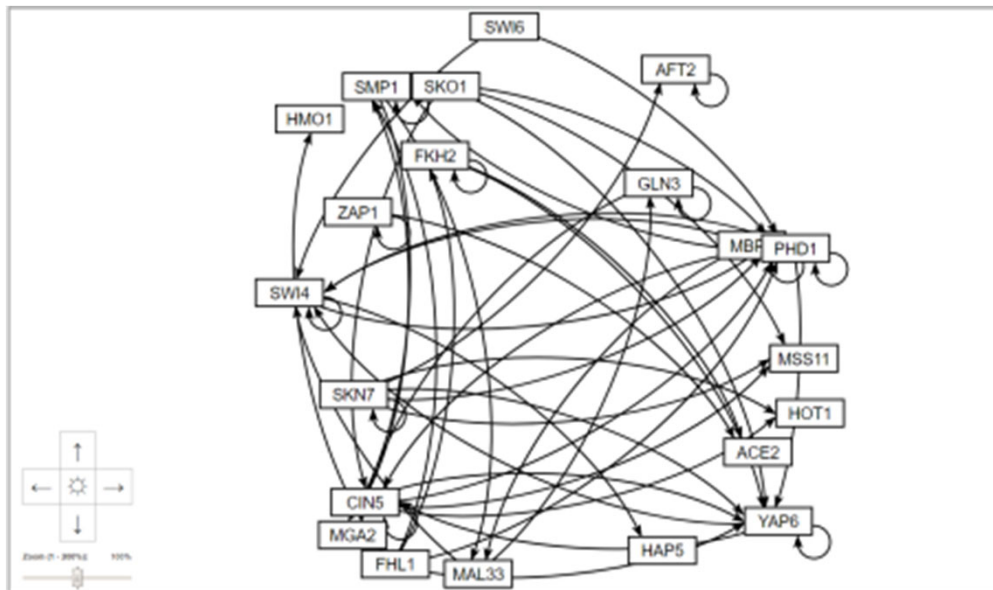
test-data2



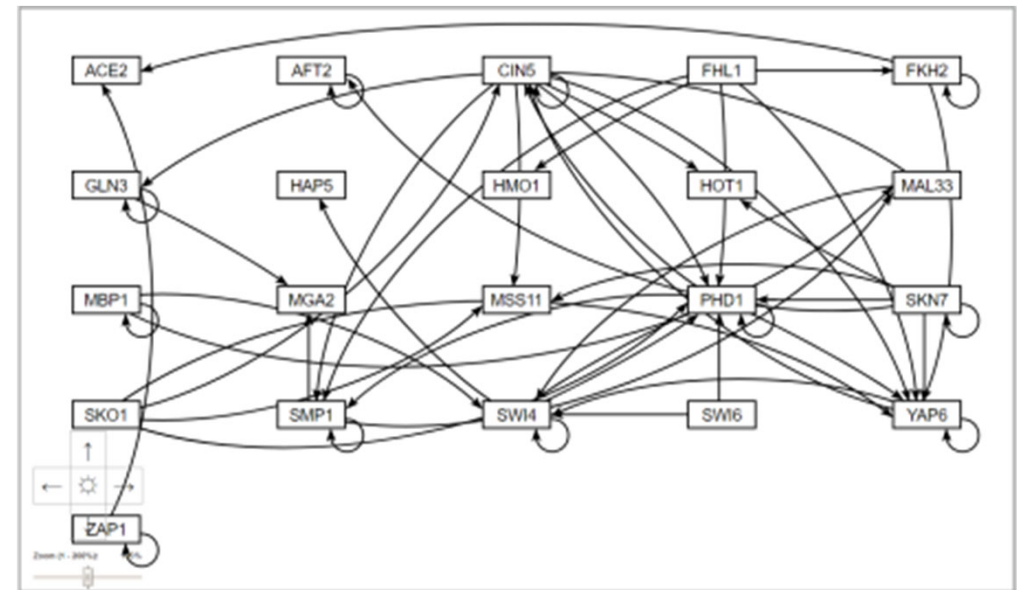
- The nodes are laid out in a grid pattern with columns and rows.
- The nodes are equally spaced out with margins around them.
- Allows users to compare and analyze data in a easier and faster way.

Sort the nodes in alphabetical order

Before Grid Layout



After Grid Layout



- The nodes are organized in alphabetical order, left to right first then top to bottom.
- Allows for easier comparison from graph to graph.

Conclusion

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GRNsight application and code are available under the open source BSD license



<http://dondi.github.io/GRNsight/>



<https://github.com/dondi/GRNsight>

Current and future work documented via Github Issues

- Organizing the menu
- Adding alternate graph layout options such as hierarchical

Acknowledgements



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GRNsight Team

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